Bogleheads

Historical and expected returns

Historical and expected returns provides historical market data as well as estimates of future market returns.

Historical returns

Burton Malkiel

Burton Malkiel, author of *A Random Walk Down Wall Street* provides historical asset class returns.

Historical (nominal) returns 1926-2005, source: Ibbotson Associates[1]

Series	Geometric mean	Arithmetic mean	Standard deviation
Large company stocks	10.4%	12.3%	20.2%
Small company stocks	12.6%	17.4%	32.9%
Long-term corporate bonds	5.9%	6.2%	8.5%
Long-term government bonds	5.3%	5.5%	5.7%
U.S. Treasury bills	3.7%	3.8%	3.1%
Inflation	3.0%	3.1%	4.3%

Robert Shiller

Robert Shiller is a Professor of Economics at Yale University. He maintains multiple historical market data spreadsheets, including the following:

- U.S. Stock Markets 1871-Present and CAPE Ratio (https://shillerdata.com/)

Professor Shiller's spreadsheet does not include historical returns, but they can be easily derived:

- Bogleheads forum post: "Computing Historical Total Returns from Shiller's data" (https://www.bogleheads.org/forum/viewtopic.php?p=7319529#p7319529)

Aswath Damodaran

Aswath Damodaran is a Professor of Finance at New York University. He maintains various corporate finance and valuation spreadsheets, including the following:

- Annual Returns on Stock, T.Bonds and T.Bills: 1928 - Current (https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histretSP.html)

Data before 1970

Forum member McQ has provided references for finding older historical data on the returns earned on stocks, bonds, and where available, other investment assets. It includes US and International data over various time frames, along with commentary on various issues related to the data.

- Bogleheads forum topic: "Historical asset returns before 1970: Where to find" (https://www.bogleheads.org/forum/viewtopic.php?t=353607), McQ. July 15, 2021
  - Bogleheads forum post: "Historical STOCK returns in the US 1926 to 1972" (https://www.bogleheads.org/forum/viewtopic.php?p=6120716#p6120716)
  - Bogleheads forum post: "Historical BOND returns in the US 1926-1970" (https://www.bogleheads.org/forum/viewtopic.php?p=6122515#p6122515)
  - Bogleheads forum post: "Historical OTHER ASSET returns in the US 1926 -1970" (https://www.bogleheads.org/forum/viewtopic.php?p=6124265#p6124265)
  - Bogleheads forum post: "Historical STOCK returns in the US 1871 to 1926" (https://www.bogleheads.org/forum/viewtopic.php?p=6126474#p6126474)
  - Bogleheads forum post: "Historical US BOND returns 1871 to 1926" (https://www.bogleheads.org/forum/viewtopic.php?p=6127990#p6127990)
  - Bogleheads forum post: "Historical OTHER ASSET returns 1871 to 1926" (https://www.bogleheads.org/forum/viewtopic.php?p=6133385#p6133385)
  - Bogleheads forum post: "Historical US STOCK returns 1793 to 1871" (https://www.bogleheads.org/forum/viewtopic.php?p=6135222#p6135222)
  - Bogleheads forum post: "Historical BOND returns 1793 to 1871" (https://www.bogleheads.org/forum/viewtopic.php?p=6137436#p6137436)
  - Bogleheads forum post: "Historical OTHER ASSETS returns 1793 to 1871" (https://www.bogleheads.org/forum/viewtopic.php?p=6142346#p6142346)
  - Bogleheads forum post: "International STOCK and BOND returns 1900 to 1972" (https://www.bogleheads.org/forum/viewtopic.php?p=6144004#p6144004)
  - Bogleheads forum post: "International STOCK and BOND returns 1600 to 1900" (https://www.bogleheads.org/forum/viewtopic.php?p=6147718#p6147718)

Expected future returns

John Bogle

John Bogle with reasonable expectations for stocks and bonds over the next ten years, using his own "Occam's razor" model.[2]

Ten-year expected long term nominal returns (2017)[3]

Asset class	Expected nominal return
U.S. stocks (broad market)	4%
U.S. bonds (broad market + moderate risk)	3.1%

Note: Bonds expected return assumes accepting moderate additional credit risk and significant interest rate risk vs. the U.S. Treasury 10-year note.

William Bernstein

William Bernstein with a summary of reasonable expected returns over the next ten years, derived from the dividend discount model (https://www.investopedia.com/terms/d/ddm.asp), published in 2014.

Ten-year expected long term real returns (2014)
Source: Rational Expectations: Asset Allocation for Investing Adults (Investing for Adults) (Volume 4)[4][5]

Asset class	Expected real return
U.S. large-cap stocks	2%
U.S. large-value and small-cap stocks	3%
U.S. small-value stocks	4%
Developed foreign stocks	5%
Emerging markets stocks	4%
REITs	1%
Precious metals stocks	1%
Base metals and oil stocks	3%
Treasury bills, notes, and bonds	-1%

William Bernstein's estimates in 2002 are below:

Ten-year expected long term real returns (2002)
Source: The Four Pillars of Investing[6]

Asset class	Expected real return
Large U.S. stocks	3.5%
Large foreign stocks	4%
Large value stocks (foreign and domestic)	5%
Small stocks (foreign and domestic)	5%
Small value stocks (foreign and domestic)	7%
Emerging market / Pacific rim stocks	6%
REITs	5%
High-yield ("junk") bonds	5%
Investment-grade corporate bonds; TIPS	3.5%
Treasury bills and notes	0-2%
Precious metals equity	3%

Rick Ferri

Rick Ferri is the author of *All About Asset Allocation*. The table below shows expected returns for all major equity and fixed income asset classes over the next thirty-years.

Thirty-year return estimate of bonds, stocks and REITs (2015)
Source: Portfolios Solutions[7]

Asset classes	Real returns	Nominal returns with 2% inflation	Risk estimate*
Government backed fixed income			
US Treasury bills (1 year maturity)	0.1	2.1	2.0
10-year U.S. Treasury notes	1.9	3.9	7.0
20-year U.S. Treasury bonds	2.5	4.5	8.0
30-year inflation protected Treasury (TIPS)	2.6	4.6	9.0
GNMA mortgages	2.4	4.4	8.0
10-year tax-free municipal (A rated)	1.6	3.6	7.0
Corporate and emerging market fixed income			
10-year investment-grade corporate (AAA-BBB)	2.6	4.6	9.0
20-year investment-grade corporate (AAA-BBB)	3.3	5.3	10.0
10-year high-yield corporate (BB-B)	4.0	6.0	15.0
Foreign government bonds (unhedged)	2.4	4.4	9.0
US equity - common equity and REITs			
US large stocks	5.0	7.0	19.0
US small cap stocks	5.3	7.3	22.0
US small value stocks	5.8	7.8	26.0
US Real Estate Investment Trusts (REITs)	4.8	6.8	19.0
International equity (unhedged)			
Developed countries	5.4	7.4	20.0
Developed countries small companies	5.7	7.7	23.0
Developed countries small value companies	6.2	8.2	27.0
All emerging markets including frontier countries	7.0	9.0	29.0

*The estimate of risk is the estimated standard deviation of annual returns.

The 30-year forecast data is presented on an annualized compounded total return basis. All interest and dividends are reinvested.

See also

- Simba's backtesting spreadsheet

External links

- Google spreadsheet (https://docs.google.com/spreadsheet/ccc?key=0AueLhNpKZa4XdFcxbzJKdnhVMDRNTTQtU2tPRXlTLVE&usp=sharing) to calculate an estimated expected return for an

entire portfolio.

References

1. Burton Malkiel (2007). *A Random Walk Down Wall Street*. p. 185. ISBN 0-393-03888-2.
2. John Bogle (2015). "Occam's Razor Redux: Establishing Reasonable Expectations for Financial Market Returns" (http://www.iijournals.com/doi/abs/10.3905/jpm.2015.42.1.119).
3. John Bogle (October 19, 2017). "Bogleheads XVI Presentation" (https://drive.google.com/file/d/0B8VSNcUR-B6Ic1VuUEY5RXptZnM/view). Slides 64 to 65.
4. Rational Expectations: Asset Allocation for Investing Adults (Investing for Adults) (Volume 4) - William Bernstein (2014)
5. Bogleheads forum topic: "Bernstein: A Decade of Super-Low Returns" (https://www.bogleheads.org/forum/viewtopic.php?t=140830). June 9, 2014
6. William Bernstein (2010). *The Four Pillars of Investing*. McGraw Hill. p. 72. ISBN 978-0-07-174705-9.
7. "Portfolio Solutions' 30-Year Market Forecast for 2015" (https://web.archive.org/web/20150407183501/https://rickferri.com/blog/investments/portfolio-solutions%C2%AE-30-year-market-forecast-for-2015/). Portfolio Solutions. March 26, 2015. Archived from the original (http://www.rickferri.com/blog/investments/portfolio-solutions%C2%AE-30-year-market-forecast-for-2015/) on April 7, 2015.. Reprinted with permission from Rick Ferri, CEO, Portfolio Solutions.

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